

U1A5 — Validation Plan and Technical Logbook

SCARA_PyBot — Embedded Systems / Robotics

1. Selected KPIs

The following key performance indicators were selected because they directly correspond to the core function of the SCARA arm (accurate, repeatable pick-and-place motion) and to the two risks identified as most critical in the U1A4 phase (simultaneous motor current draw and Wi-Fi command reliability):

- Positioning accuracy — does the arm reach the coordinate it was commanded to reach?
- Repeatability (homing error) — is the zero reference stable across repeated homing cycles?
- Power consumption — does the 12 V/5 A supply support all 3 motors moving at once?
- Wi-Fi command latency — how quickly does the arm react to a remote command?
- Step pulse rate / motion stability — is the generated STEP signal accurate and jitter-free?

2. Validation Table

KPI	What is measured	Unit	Instrument / Tool	Measurement procedure	Expected value	Acceptance criterion
Positioning accuracy	Deviation between the commanded end-effector coordinate (X,Y,Z) and its actual physical position	mm	Digital caliper + fixed reference jig / printed grid under the work plane	Command the arm to 5 known reference points via the USB/Wi-Fi API; mark the actual tip position; measure the Euclidean deviation from the target with the caliper	≤ 2 mm average deviation	≤ 3 mm maximum deviation at any of the 5 points
Repeatability (homing error)	Variation of the end-effector position after repeating the homing (zero-reference) routine	mm (std. dev.)	Digital caliper / dial indicator	Execute the homing sequence 10 consecutive times; measure the tip position after each run; compute the standard deviation of the 10 measurements	≤ 0.5 mm std. dev.	≤ 1 mm std. dev. across 10 runs
Power consumption	Current drawn from the 12 V rail in idle state and during simultaneous 3-axis motion	A / W	Multimeter (DC current clamp or in-line ammeter) on the 12 V supply line	Measure current with the system idle; then command all 3 axes to move simultaneously to opposite travel extremes and log peak and average current	Peak ≤ 3.5 A	Peak current must stay < 4 A (80% of the 5 A supply rating) with no brownout

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Wi-Fi command latency	Time between a motion command being sent from the client (web app) and the MCU starting the STEP pulse train	ms	Logic analyzer / oscilloscope on the STEP GPIO pin, synchronized with a client-side timestamp log	Send 20 sequential move commands over Wi-Fi; timestamp the send event on the client and the first STEP edge on the MCU; compute the latency of each sample	≤ 100 ms average	Average latency ≤ 150 ms and no single sample > 300 ms
Step pulse rate / motion stability	Frequency and jitter of the STEP pulse train generated for a constant-speed move	Hz (steps/s)	Oscilloscope / logic analyzer on the STEP GPIO pin	Command a constant-speed linear move on one axis; capture the STEP pulse train; measure pulse frequency and pulse-to-pulse jitter	Frequency within $\pm 5\%$ of the commanded value	Jitter $\leq 10\%$ of the nominal pulse period; no missed or doubled pulses

3. Test Cases

Test Case 1 — Homing Repeatability

Objective	Verify that the homing routine consistently returns all three axes to the same mechanical reference position.
Initial conditions	System powered on; firmware with the homing routine flashed; all axes manually displaced away from their limit switches before each run.
Procedure	1) Power on the system. 2) Trigger the homing command over USB serial. 3) Wait for the “homing complete” status message. 4) Measure the end-effector tip position with a caliper against a fixed reference point. 5) Repeat steps 1–4 ten times.
Expected result	The end-effector returns to within 1 mm of the same reference point on every run.
Pass / Fail criterion	PASS if the standard deviation of the 10 measurements is ≤ 1 mm; FAIL otherwise.

Test Case 2 — Point-to-Point Positioning Accuracy

Objective	Validate that commanded (X, Y) coordinates match the actual physical position of the end effector within tolerance.
Initial conditions	Homing completed successfully; a grid sheet with 5 marked reference points fixed under the work plane.
Procedure	1) Send a move command to reference point 1 via the serial/Wi-Fi API. 2) Measure the actual tip position against the marked point with a caliper. 3) Record the deviation. 4) Repeat for the remaining 4 points.
Expected result	Deviation ≤ 3 mm at every one of the 5 points.
Pass / Fail criterion	PASS if all 5 points show ≤ 3 mm deviation; FAIL if any point exceeds 3 mm.

Test Case 3 — Power Draw Under Full 3-Axis Load

Objective	Confirm the 12 V/5 A power supply and the A4988 drivers support simultaneous motion of all three axes without a brownout or step loss.
Initial conditions	Ammeter/multimeter connected in series with the 12 V input; VREF on all 3 A4988 drivers set to the rated current.
Procedure	1) Command a simultaneous move of all 3 axes toward opposite travel extremes. 2) Monitor current draw continuously during the move. 3) Record peak and average current. 4) Observe for MCU resets or audible step loss.
Expected result	Peak current stays below 4 A; no MCU reset; no missed steps.
Pass / Fail criterion	PASS if peak current < 4 A and no reset/step-loss is observed; FAIL otherwise.

Test Case 4 — Wi-Fi Command Latency and Connection-Loss Safety Stop

Objective	Verify that remote command latency is acceptable and that the firmware safety timeout stops all motors if the Wi-Fi connection is lost mid-motion.
Initial conditions	ESP32-S3 connected to the Wi-Fi network; client application running on a PC/phone on the same network; safety timeout configured in firmware.
Procedure	1) Send 20 sequential move commands and log latency for each. 2) During an active motion, disconnect the Wi-Fi client. 3) Observe firmware behavior after the timeout period elapses.
Expected result	Average latency ≤ 150 ms; motors stop automatically within the configured timeout (e.g., 2 s) after connection loss.
Pass / Fail criterion	PASS if the latency criterion is met and the motors halt within the timeout window; FAIL otherwise.

4. Technical Logbook — Entry 1

Date	September 14, 2026
Team members	Jassiel Obed Jiménez Ochoa (update this field with teammates' names if the project is completed as a team of up to 4)
Current project status	Phase 1 (system architecture, main components, HW/SW partition and preliminary BOM, U1A4) is complete. Phase 2 (validation planning) is in progress: KPIs, validation table and test cases defined in this document.
Work completed	- Reviewed the U1A4 architecture, block diagram and BOM as the baseline for validation. - Selected 5 KPIs relevant to the SCARA arm (positioning accuracy, repeatability, power consumption, Wi-Fi latency, step-pulse stability). - Built the validation table with instruments, procedures, expected values and acceptance criteria. - Defined 4 reproducible test cases covering homing, accuracy, power and communication safety. - Created the project repository skeleton (hardware/, firmware/, documentation/, tests/, README.md).
Problems found	- No oscilloscope/logic analyzer has been reserved yet for the STEP-pulse and Wi-Fi latency tests (TC4 and the step-rate KPI); this needs to be scheduled with the lab. - The A4988/ESP32-S3 lead time risk identified in U1A4 still stands — components have not been ordered yet, which could delay the first physical test run.
Decisions made	- Positioning accuracy was chosen as the primary KPI, since it most directly reflects the pick-and-place purpose of the arm. - Homing repeatability will be validated before absolute positioning accuracy, since the zero reference must be trustworthy first. - A simple caliper-and-grid method was chosen over a vision-based measurement system for the first validation pass, to keep the test setup low-cost and fast to execute.

Next task	Order the BOM components, set up the ESP32-S3 firmware environment (FastAccelStepper bench test per U1A4 risk #5), and run Test Case 1 (homing repeatability) as soon as the mechanical assembly and at least one axis are wired.
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5. Project Repository

The project repository “SCARA_PyBot” was created locally with the following structure, ready to be pushed to a remote Git host (e.g., GitHub):

- SCARA_PyBot/hardware/ — BOM, block diagram and wiring references (from U1A4)
- SCARA_PyBot/firmware/ — ESP32-S3 source code (kinematics, motion control, homing, Wi-Fi API)
- SCARA_PyBot/documentation/ — U1A4 report, this validation plan, and the running technical logbook
- SCARA_PyBot/tests/ — test case records and captured results (oscilloscope captures, latency logs, accuracy measurements)
- SCARA_PyBot/README.md — project overview

Repository link: [to be added once pushed to the team's chosen Git host — see README.md for the local structure in the meantime].

Analysis Questions

1. Which KPI is the most important for your project and why?

Positioning accuracy is the most important KPI, because the core function of the SCARA arm is to place the end effector at a specific coordinate to pick and place objects; every other KPI (power consumption, latency, step-pulse stability) exists to support or protect this primary function rather than define it. Repeatability of the homing routine is a close second, since positioning accuracy is meaningless if the zero reference itself is not trustworthy.

2. What instrument will be required to measure it?

A digital caliper together with a fixed mechanical reference (a jig or a printed grid under the work plane) is sufficient to measure positioning accuracy at the tolerances required for this prototype (millimeter range). A camera-based or vision-guided measurement system would give more data points but is not required at this validation stage.

3. What result would indicate that the prototype does not meet the requirements?

A measured positional deviation greater than 3 mm at any of the 5 reference points, or a homing standard deviation above 1 mm across repeated runs, would indicate the prototype fails the accuracy requirement. Independently, an MCU reset or audible step loss during the full 3-axis load test would indicate the power design does not meet the requirement, regardless of how accurate the arm is when only one axis moves at a time.

4. Which test should be performed first?

Test Case 1 (homing repeatability) should be performed first, because every other test — positioning accuracy, and indirectly the power and latency tests, which are also referenced to a known position — assumes that the zero reference produced by homing is stable and repeatable. Validating homing first isolates that assumption before it is built into the other tests.

5. Why is a technical logbook useful during an engineering project?

A technical logbook creates a dated, traceable record of what was done, what failed, and why specific decisions were made, which is essential once component substitutions, firmware changes or test results need to be justified later — to a professor, a teammate joining midway, or the team's own future self. It also

turns implicit knowledge (e.g., why a given acceptance criterion or instrument was chosen) into an explicit record that keeps the whole team, not just one member, able to reproduce and continue the validation work.